

CSCE 632 ACCESSIBLE COMPUTING · MODULE 06 · LECTURE NOTES

Cognition, Attention, and Language

The largest group, and the least served.

MODULE OUTCOMES

- Explain why this area is the least well covered.
- State the memory and attention requirements.
- Explain language and reading requirements.
- Explain error prevention and recovery as access.
- Reduce the cognitive demand of an interface.

1 Why this is underserved

THE COGNITIVE REQUIREMENTS ARE THE LARGEST IN POPULATION AND THE WEAKEST IN THE STANDARD

Cognitive, learning, and attention-related conditions are collectively the most common category of disability — and the standard's coverage of them is the thinnest and the most qualitative of any area it addresses, which is a mismatch worth naming.

Because the requirements resist the measured-not-judged property that makes the contrast criterion so effective (Module 03 §3): **'is this simple enough' has no threshold, and a criterion without a threshold is very hard to enforce** — so the standard either omits it or states it at a level nobody is required to meet.

Which is a limitation of the standard rather than of the requirement — and Module 07 §4 is about exactly that distinction: what a standard can encode, and what it therefore leaves to you.

So this module is more design guidance than checklist, stated openly, because **pretending a qualitative requirement is a checkbox produces compliance without benefit** — which is the characteristic failure of accessibility programmes and Module 12 §3's subject.

2 Memory and attention

- Do not require remembering across steps — show the reference code on the page that asks for it, show the email address on the page that asks you to confirm it — **which is CSCE 671 Module 03's recognition-over-recall principle** as an access requirement.
- Do not require remembering across modes — a modal dialogue that covers the information it is asking about is the specific and extremely common version of this, and it is a layout decision rather than a hard problem.

- **Keep the sequence short and show the position within it** — 'step 2 of 4' costs nothing and removes a **sustained background uncertainty** about how much is left, which consumes attention continuously.
- **Allow interruption and resumption** — **which is the requirement attention conditions most need and almost nothing provides: save the partial state**, so that leaving and returning is not starting again. This is also §4's data-preservation requirement.
- **And remove involuntary attention capture** — **motion, autoplay, carousels, notification badges, and sudden layout shift all take attention whether or not it can be spared** (CSCE 671 Module 06 and CSCE 679 Module 02 §3's emphasis budget) — and the user cannot choose not to notice them.

3 Language

WRITE FOR THE LOWEST READING DEMAND THE CONTENT ALLOWS, WHICH IS LOWER THAN YOU THINK

Short sentences, common words, one idea per sentence, and the point before the explanation — which helps readers with dyslexia, with cognitive conditions, reading in a second language, and anybody reading under time pressure, which over a population is nearly everybody at some point.

And expand abbreviations and define jargon on first use — which is a mechanical requirement rather than a judgement, and is therefore checkable, unlike most of this module.

Plus declare the language of the page and of any passage in another language — because a screen reader selects its pronunciation rules from that declaration and gets it audibly and sometimes incomprehensibly wrong without it, which is a one-attribute fix for a total failure.

And this is not dumbing down: precise technical content can be written plainly, and the difficulty of the prose is independent of the difficulty of the idea — which is a claim this program's own writing tries to honour, and which you are entitled to check on it.

4 Errors

PREVENT

constrain the input instead of validating it
format the field as the user types
accept every reasonable form of the answer
and do not ask for what you already know

IDENTIFY

say which field, in text, next to the field
say what is wrong, specifically
and not only in colour (Module 03 section 3)

RECOVER

say how to fix it, not only that it is wrong
preserve everything already entered
allow review before anything irreversible
and allow reversal afterwards where possible

WHICH IS CSCE 671 SECTION 05's error material,
arriving as an access requirement -- because a
recoverable error costs a moment and an unrecoverable
one ends the task.

AND THE HONEST CLOSING POSITION

Everything in this module benefits everybody, measurably and immediately — which makes it the strongest curb cut in the course (Module 01 §2) and also, perversely, the reason it is treated as optional polish rather than as an access requirement with a population behind it.

And the gap between the size of the population and the standard's coverage is the largest in the subject, which means **a conformance claim is least informative here** (Module 07 §4) — a fully conformant interface may still be cognitively unusable, and nothing in the claim says otherwise.

So the test is not a checklist but a count: how many things must the user hold in mind, how many steps are there, how many words, how many unrecoverable actions — and every one of those is reducible, which makes the count actionable even without a threshold.

Which makes this the one module where 'make it simpler' is the whole requirement — and where your own fluency with your own interface is the thing most likely to mislead you, because you cannot experience it as unfamiliar (Module 01 §4's shared error, and CSCE 671 Module 01 §2).

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
The W3C Cognitive Accessibility materials (free)	The whole module — the task force's guidance, which is more detailed than the standard itself and is the right source here.
Making Content Usable for People with Cognitive and Learning Disabilities (free)	§§2 to 4 in depth , with patterns and the user-need statements behind them.
The UK Government Digital Service content guidance (free)	§3 as working practice — plain writing applied to genuinely complex content, which is the proof that it is not dumbing down.
Understanding WCAG 3.3.x on input assistance (free)	§4's criteria — error identification, suggestion, and prevention for consequential actions.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **State why this area resists the measured-not-judged property**, and what follows.
- 2 **Find one place your interface requires remembering across steps**, and fix it.
- 3 **Find a modal that hides what it asks about**.
- 4 **Add a position indicator** to one multi-step flow.

- 5 **Make one flow interruptible**, with its partial state saved.
- 6 **List every involuntary attention capture** in one interface you use.
- 7 **Rewrite one page of your own technical writing** for the lowest reading demand it allows.
- 8 **Check the language declarations** on a multilingual page, and listen to it.
- 9 **Audit one form against prevent-identify-recover**, all three.
- 10 **Count the unrecoverable actions** in your interface, and make one reversible.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why is this area the largest and least covered?
- 2 What property do these requirements resist, and why does that matter?
- 3 What does pretending they are checkboxes produce?
- 4 Give five ways to reduce memory and attention demand.
- 5 Which is most needed and least provided?
- 6 State the language requirements, including the technical one.
- 7 Why is plain writing not dumbing down?
- 8 Give the three error stages and two items under each.
- 9 Why is error design an access requirement?
- 10 What is the test here, if not a checklist?